

Online Traveling Salesman Games: Core Fragility and the Temporal Nucleolus under Dynamic Arrivals

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Abstract

Urban last-mile logistics—same-day delivery, ride-hailing, crowdsourced pickup—requires that the cost of a single shared tour be divided among customers whose identities are revealed only as the vehicle dispatches. Static cooperative routing models silently assume every customer is available from the outset, distorting fairness when arrivals are spread in time. We introduce the Online Traveling Salesman Game (Online TSG), in which the family of feasible coalitions is endogenously induced by the realized arrival-service process; the classical static game is recovered at simultaneous arrivals. Our main theoretical result identifies three complementary sufficient conditions under which the Core of this restricted game empties—a single-complement threshold tied to the minimum residual customer cost, a balanced-complement threshold of Bondareva–Shapley type, and a balanced-near-complement threshold on size- $(n-1)$ and $(n-2)$ coalitions—and proves that all three are structurally blocked whenever the peak waiting queue satisfies $k < n - 1$ (intermediate-coalition emptiness can persist in this regime; Observation 15). Cooperative cost allocations on this restricted family are computed via a sequential LP cascade (the Temporal Nucleolus). The single-complement threshold tightens at the Beardwood–Halton–Hammersley $\sqrt{n}$ rate as n grows. On 175 nearest-neighbor instances 67 Cores are empty; three complement-family thresholds certify 57 and a partition-pair certificate the remaining 9 (66 of 67, no false positives). Extending to cheapest-insertion and batch re-optimization (525 policy-instance pairs) yields no false positives and eliminates an intermediate-coalition mechanism observed only under nearest-neighbor. Scale-up to 50 customers confirms the asymptotic tightening. In our synthetic single-vehicle experiments, short service-time promises (peak queue $k < n - 1$) structurally block all complement-coalition mechanisms, and combined with cheapest-insertion or batch re-optimization dispatch they substantially improve empirical Core stability.

Keywords: Routing; Cooperative games; Nucleolus; Core stability; Dynamic arrivals.

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1 Introduction

Urban last-mile logistics is characterized by requests that enter the system *after* planning begins: online grocery orders, same-day parcels, ride-hail queues and crowdsourced pickups all share the feature that the identity of who will be served together with whom is not known until dispatch time (Pillac et al., 2013; Ausiello et al., 2001). Operators nevertheless must share the routing cost among the customers who are eventually served on a joint tour, and the quality of this allocation shapes incentives for participation and pricing (Guajardo & Rönnqvist, 2016). Cooperative game theory, and specifically the Traveling Salesman Game (TSG) of Potters et al. (1992), supplies the canonical framework in which a tour cost is divided so that no subset of customers has a strict incentive to defect; the Vehicle Routing Game (VRG) of Göthe-Lundgren et al. (1996); Engevall et al. (2004) extends the idea to multi-vehicle settings. Classic applications of the VRG include school bus routing (Park et al., 2012), where the set of customers is fixed before the vehicle departs. Online counterparts arise naturally in same-day delivery and ride-hailing platforms, where requests arrive dynamically and the coalition of customers to be served jointly is revealed only during execution.

The classical TSG, however, is defined under the tacit assumption that every customer is available to be visited at any point of the tour: any coalition $S \subseteq N$ can be *imagined* as a self-contained instance, and its characteristic cost $c(S)$ is the optimal TSP tour on $S \cup \{0\}$. In online settings, this counterfactual is not available: a coalition S is operationally meaningful only if there exists a moment at which all members of S have arrived but none have yet been served. Ignoring temporality hides a central source of instability. Two customers who arrive far apart in time are never candidates to be served together, and pretending that they form a coalition distorts fairness. Recent work on online cooperative games (Ge et al., 2024; Aziz et al., 2025; Goyal et al., 2025) has begun to address temporality but not in the routing context.

Research question. What is the correct cooperative game-theoretic abstraction of the online traveling salesman problem, how does it compute, and—most importantly—when does it admit a stable allocation?

Contributions. We make four contributions.

- C1. Model.** We define the Online TSG $(N, c, \mathcal{F})$ by restricting coalitional stability constraints to the family $\mathcal{F}$ of proper sub-coalitions realizable under the realized arrival dynamics, with the grand coalition handled through the efficiency equality. This places the Online TSG within the taxonomy of restricted-cooperation games (Section 2) but with a novel source of restriction: $\mathcal{F}$ is not an exogenous graph or combinatorial structure but an endogenous random family induced by the arrival-service process. The classical static TSG is recovered as the special case where arrivals collapse to a single instant and the realized tour is optimal (Supplementary Proposition 2, §S6). Cooperative cost allocations on this restricted family are computed via a sequential LP cascade (the Temporal Nucleolus, Section 4).

- C2. Core theory under dynamics.** We provide three complement-family sufficient conditions for empty Core (Theorem 9 $r > r^{**}$, single-complement; Proposition 10 $r > r^{***}$, balanced-complement on $B_{n-1} = \{N \setminus \{i\}\}$; Proposition 13 $r > r^{(\diamond)}$, balanced-near-complement on the mixed collection $B_{n-1} \cup B_{n-2}$) plus one unifying partition-pair certificate (Proposition 14 $r > r_{\min}^{(P)}$) that contains Theorem 9 as the singleton-complement case and additionally fires on non-singleton partition pairs at $k < n - 1$. Corollary 18 provides a structural obstruction for the three complement-family conditions; Proposition 14 survives at $k < n - 1$ and certifies the 9 cases of Observation 15 that complement-only analyses leave open.
- C3. Asymptotic refinement.** Under uniform coordinate sampling, $r^{**} \rightarrow 1$ at rate $O(n^{-1/2})$ almost surely (Theorem 16); this explains the empirical smallness of r^{**} without reference to specific simulation parameters and ties the cooperative-game threshold to the classical Beardwood–Halton–Hammersley $\sqrt{n}$ scaling of the Euclidean TSP.
- C4. Empirical calibration.** On 175 simulated instances across five coalition sizes and seven arrival patterns under three dispatch policies (525 policy-instance pairs), plus 45 scale-up instances up to $n = 50$ and a 24-instance scale-invariance study (Supplementary Materials, §S2), we verify every theoretical claim. Under NN dispatch, Theorem 9 fires in 37 instances (all observed empty), Proposition 10 adds 11 balanced-complement certifications, and Proposition 13 adds 9 balanced-near-complement certifications (jointly 57 of 67 observed empties); the remaining 10 split as 1 near-complement residual (threshold-bound by $\approx 2 \times 10^{-3}$) and 9 intermediate-coalition cases certified by Proposition 14 (66 of 67 total). No false positives arise under any policy. The mean thresholds ($\bar{r}^{**} = 1.223$, $\bar{r}^{***} = 1.096$) are sharper than the classical bound ($\bar{r}^* = 4.351$) by factors of 3.56 and 3.97. Corollary 18 holds analytically without exception: no instance with $k < n - 1$ satisfies any of the three thresholds; the 9 intermediate-coalition empties all lie at $k < n - 1$ under NN and vanish under cheapest-insertion and batch-reoptimization. The asymptotic decay $r^{**} \rightarrow 1$ is consistent with Theorem 16’s $O(n^{-1/2})$ rate on the scale-up grid.

Operationally these results say: a bounded peak queue $k < n - 1$ —achievable via short service-time promises—eliminates all complement-coalition mechanisms (Theorem 9, Propositions 10 and 13), covering the pathological “morning batch” regime. In the shallow-queue regime the intermediate-coalition mechanism of Observation 15 remains active but is policy-sensitive: observed only under nearest-neighbor dispatch and absent under cheapest-insertion or batch re-optimization (Section 6.6). Short service windows combined with moderately informed dispatch substantially improve empirical Core stability, though intermediate-coalition emptiness remains logically possible at high competitive ratios.

Structure. The Online TSG is a single-vehicle cooperative game; this scope isolates the temporal coalition-feasibility structure of $\mathcal{F}$ from fleet-assignment confounds and is discussed further in Section 7 (where a multi-vehicle Online VRG is deferred to a companion manuscript). Section 2

situates our work within the literature on static TSG/VRG, dynamic and rolling-horizon games, online cooperative games, and nucleolus computation. Section 3 formalizes the Online TSG. Section 4 defines and computes the Temporal Nucleolus. Section 5 develops the Core theory. Section 6 reports computational experiments, and Section 7 concludes.

2 Related Work

We organize prior work along three streams and state, for each, the precise point of departure of our contribution.

2.1 Cooperative Routing Games: Static, Dynamic, and Rolling-Horizon

The cooperative TSP was initiated by Potters et al. (1992) with a characteristic function equal to the optimal TSP tour cost on the chosen coalition. Tamir (1989) showed that the Core may be empty even on the Euclidean line, and the gap between fair and feasible allocations was quantified via approximation in Faigle et al. (1998) and Bläser & Shankar Ram (2008). In parallel the Vehicle Routing Game (VRG) received a Nucleolus-based treatment by Göthe-Lundgren et al. (1996), extended to heterogeneous fleets by Engevall et al. (2004) and to time-window settings by Tae et al. (2020) and Arroyo (2024). Most recently, Lyu et al. (2025) study the collaborative berth allocation problem and compute stable cost allocations via row generation; their coalition family is static and determined by vessel identity rather than arrival dynamics, which is the source of restriction that differs from our setting. Platz & Hamers (2015) extend the routing-plus-cooperative-game template to multi-depot Chinese postman problems, but with coalition family 2^N rather than a time-restricted $\mathcal{F}$. The Shapley value formulation of cooperative TSP under rolling horizons is treated by Kimms & Kozeletskyi (2016b), and a core-based variant by Kimms & Kozeletskyi (2016a); a broad survey is given by Guajardo & Rönnqvist (2016). Özener & Ergun (2008) treat operational cost allocation among carriers in a collaborative transportation procurement network with a fixed carrier set. Closest in spirit to our work, Özener et al. (2013) combine routing, cost allocation, and a temporal dimension in inventory routing; their temporality is a deterministic replenishment schedule rather than an online stochastic arrival process restricting $\mathcal{F}$. On the temporal side, Kranich et al. (2005) proposed Core concepts for TU games played over discrete time, Habis & Herings (2010) defined the weak sequential Core to accommodate renegotiation, and Lehrer & Scarsini (2013) studied how the Core evolves as information unfolds. *Departure*. Every cited study either defines coalitions over the full customer set 2^N (so coalition identity is assumed known *a priori*) or abstracts away the physical routing cost structure, describing coalitions combinatorially with exogenous values $v(S, t)$. We replace 2^N with the data-dependent family $\mathcal{F}$ of coalitions realizable under the online arrival process while retaining the metric TSP cost $c(S)$, so temporality enters through the *admissibility* of coalitions and couples the game to the dispatch policy.

2.2 Restricted Cooperation Games

A parallel line of cooperative game theory, independent of the routing literature, studies games in which the set of formable coalitions is a *strict subset* of 2^N . Myerson (1977) initiated this theme by treating coalitions as paths in an underlying communication graph, producing the graph-restricted game (N, v^c) in which $v^c(S)$ aggregates values of S 's connected components. Faigle (1989) generalized the framework to games on combinatorial structures (antimatroids, convex geometries, partitions), and Bilbao (2000) provided a unified treatment. Algaba et al. (2001) connected Myerson's and Faigle's lines. More recent extensions handle permission structures (Gilles, 2010), hierarchical coalition configurations, and the core of games on ordered structures (Grabisch, 2013). Computing the nucleolus of a restricted game is in general no harder than the unrestricted case and may be substantially easier, since the constraint count is $|\mathcal{F}|$ rather than $2^n - 1$ (Schouten et al., 2022). *Departure.* In this restricted-cooperation literature, feasibility is determined by an *exogenous and time-invariant* structure—a graph, a partition, an antimatroid, or a permission relation—specified as part of the game's primitives. The Online TSG introduces a different source of restriction: feasibility is *endogenously induced* by the realized dynamics of a stochastic arrival-service process. Formally, $\mathcal{F}$ is a data-dependent random family with $S \in \mathcal{F}$ iff customers in S share a nonempty common waiting window $\text{CW}(S) = \bigcap_{i \in S} [a_i, s_i]$ (Definition 2). The resulting family $\mathcal{F}$ is closed under neither union nor intersection in general, and its shape depends on the arrival-order permutation, the queue-depth process, and the dispatch policy. Our theoretical contributions—the complement-coalition threshold r^{**} (Theorem 9), the structural obstruction at bounded peak queue (Corollary 18), and the asymptotic tightening (Theorem 16)—are specific to this time-induced restriction and have no direct counterpart in the graph-restricted or antimatroid literatures. We view the Online TSG as adding *temporally-induced restricted cooperation* to the taxonomy of restricted-cooperation games.

A separate line of theoretical work relevant to our asymptotic analysis (Theorem 16) is the probabilistic theory of Euclidean combinatorial optimization, initiated by Beardwood et al. (1959) and systematized by Steele (1981, 1997). The Beardwood–Halton–Hammersley theorem characterizes the $\sqrt{n}$ scaling of the optimal Euclidean TSP tour, which in our setting controls the denominator of r^{**} and hence its asymptotic tightening. To the best of our knowledge, this asymptotic perspective on cooperative routing games is new: prior cooperative TSP and VRG studies (Potters et al., 1992; Göthe-Lundgren et al., 1996; Engevall et al., 2004; Kimms & Kozeletskyi, 2016a) treat the coalition cost as a fixed quantity rather than as a random variable with a known limit distribution.

2.3 Online Cooperative Games and Nucleolus Computation

Bullinger & Romen (2023) studied online coalition formation under random arrival and dissolution. Ge et al. (2024) and Zhang et al. (2025) designed incentives for early arrival in online cost-sharing settings. Aziz et al. (2025) characterized participation incentives for players entering over time. Goyal et al. (2025) introduce Temporal Cooperative Games, in which the characteristic function is indexed by time. Their worth $v(\pi)$ depends on the arrival sequence while coalitions

remain unrestricted; our $c(S)$ is set-valued but coalitions are restricted to $\mathcal{F}$, an orthogonal axis of temporality. On the algorithmic side, [Schmeidler \(1969\)](#) introduced the Nucleolus as the lexicographic minimizer of the excess vector; the sequential-LP approach that we adopt descends from Kohlberg’s algorithm and its modernization by [Guajardo & Jörnsten \(2015\)](#), with further scaling to large games by [Nguyen & Thomas \(2016\)](#), Nucleolus-based ridesharing in [Lu & Quadrioglio \(2019\)](#), Nucleolus computation for the VRG with time windows (VRPTW) in [Tae et al. \(2020\)](#), and a Core-targeted LP methodology for cooperative procurement in [Drechsel & Kimms \(2010\)](#) that serves as an early precedent for the constraint-generation style we adapt to $\mathcal{F}$. [van Zon et al. \(2021\)](#) compute Core allocations for the Joint Network Vehicle Routing Game via a row-generation algorithm over a static coalition family; our sequential-LP cascade over the time-induced family $\mathcal{F}$ is methodologically analogous but applies to a different, dynamically restricted coalition structure. *Departure.* The online-game literature differs from ours in two respects. First, [Bullinger & Romen \(2023\)](#) formalize coalition *formation* (which players group with whom) rather than cost sharing within a fixed grand coalition, so their stability concepts and ours address distinct strategic problems. Second, and common to the whole stream, none of these models equip the online coalition process with a TSP or routing cost structure; the inequalities that define our Temporal Core reflect exactly the geometric interactions among customers in the plane, allowing the sharp thresholds r^{**} and structural bound $k < n - 1$ established in [Section 5](#). On the algorithmic side, our LP cascade is defined over $\mathcal{F}$ rather than $2^N \setminus \{\emptyset\}$: in sequential arrival regimes this can reduce the per-LP constraint count by more than an order of magnitude ([Section 6](#)).

3 Problem Definition

3.1 Notation

We consider a depot 0 and n customers $N = \{1, \dots, n\}$, located in a metric space with symmetric travel cost d . A single vehicle leaves the depot, visits a subset of customers, and returns. Each customer $i \in N$ is associated with a random *arrival time* $a_i \in \mathbb{R}_+$, revealed online, at which its request enters the system, and a *service time* $s_i \geq a_i$ at which the vehicle first serves it in the realized execution. [Table 1](#) collects the notation used throughout.

3.2 Feasible coalitions and the characteristic function

The starting point is that the characteristic cost $c(S)$ of a coalition is *defined in the same way as in the static TSG*: it is the cost of the optimal TSP tour on $S \cup \{0\}$ irrespective of time. Temporality enters only through the admissibility of coalitions.

Definition 1 (Common waiting window). For a coalition $S \subseteq N$, $\text{CW}(S) := \bigcap_{i \in S} [a_i, s_i]$.

Definition 2 (Feasible coalition). A nonempty coalition $S \subseteq N$ is *feasible* under the realized

Table 1: Notation.

Symbol	Meaning
N, n	Grand coalition of customers and its cardinality
$d(i, j)$	Symmetric travel cost (metric) from i to j
$c(S)$	Optimal TSP tour cost on $S \cup \{0\}$, <i>time-independent</i>
$c^*(N)$	$c(N)$; the grand-coalition TSP optimum
a_i, s_i	Arrival and service time of customer i
U_t	Customers that have arrived by t and are not yet served, $U_t = \{i : a_i \leq t < s_i\}$
$\mathcal{F}$	Family of <i>feasible coalitions</i> : $S \in \mathcal{F}$ iff $ S = 1$ or $\max_{i \in S} a_i < \min_{i \in S} s_i$ (Definition 2)
$\text{CW}(S)$	Common window $\bigcap_{i \in S} [a_i, s_i]$
$C(N)_{\text{online}}$	Cost of the online tour realized by the dispatch policy
r	Competitive ratio $C(N)_{\text{online}}/c^*(N)$
δ_i	$c(\{i\}) + c(N \setminus \{i\}) - c^*(N)$
r^{**}	$1 + \min_{i: N \setminus \{i\} \in \mathcal{F}} \delta_i/c^*(N)$: tight threshold (Thm. 9); undefined if no such i exists
r^{***}	$1 + (\sum_{i \in N} \delta_i)/((n-1)c^*(N))$: balanced-complement threshold (Prop. 10); requires $B_{n-1} \subseteq \mathcal{F}$
$r_{\min}^{(\diamond)}$	LP optimum of Bondareva–Shapley over $B_{n-1} \cup B_{n-2}$ (Prop. 13); requires $B_{n-1} \cup B_{n-2} \subseteq \mathcal{F}$
r^*	$\sum_i c(\{i\})/c^*(N)$: Remark 5 upper bound
k	$\max_t U_t $: peak size of the waiting queue

arrival/service sequence (a, s) if either $|S| = 1$, or $|S| \geq 2$ and $\max_{i \in S} a_i < \min_{i \in S} s_i$. Let

$$\mathcal{F} := \{S \subseteq N : S \neq \emptyset, |S| = 1 \text{ or } \max_{i \in S} a_i < \min_{i \in S} s_i\}$$

denote the family of feasible coalitions. Equivalently, $S \in \mathcal{F}$ iff $\text{CW}(S)$ is nonempty *and* has nonempty interior whenever $|S| \geq 2$, i.e. iff there exists t with $a_i \leq t < s_i$ for every $i \in S$ or S is a singleton.

Under the closed-interval convention of Definition 1, $\text{CW}(\{i\}) = [a_i, s_i] \ni a_i$ is automatically nonempty for every $i \in N$, even when the vehicle reaches i at its arrival instant (i.e. $s_i = a_i$), so singletons are always feasible without an auxiliary convention. For $|S| \geq 2$, the strict inequality $\max_{i \in S} a_i < \min_{i \in S} s_i$ used in Supplementary Lemma 4 and in our implementation identifies the open interior of $\text{CW}(S)$ within which every member strictly co-waits before any service begins; boundary configurations with $\max a_i = \min s_i$ have Lebesgue measure zero under continuous arrival distributions and correspond to a member arriving at the instant another member’s service begins, and are excluded on operational grounds. The grand coalition $N \in \mathcal{F}$ whenever all n customers are queued simultaneously. We write $\mathcal{F}^p := \mathcal{F} \setminus \{N\}$ for the family of *proper* feasible coalitions; the Temporal Core and Temporal Nucleolus are defined on $\mathcal{F}^p$, with N entering only through the efficiency equality (Myerson, 1977; Bilbao, 2000).

Ex-post, policy-induced character of $\mathcal{F}$. The family $\mathcal{F}$ is determined ex post by the realized arrival-service sequence (a, s) , where s is the realized service time under the deployed dispatch policy; the Online TSG Core is therefore an ex-post, policy-induced stability concept rather than a counterfactual one. Sensitivity of $\mathcal{F}$ to the choice of dispatch policy is analyzed in Section 7 (Limitation (v)).

Table 2: Static TSG versus Online TSG.

	Static TSG (Potters et al., 1992)	Online TSG (this paper)
Coalitions considered	All $S \subseteq N$, $S \neq \emptyset$	$S \in \mathcal{F}$
Characteristic function	$c(S) = \text{TSP on } S \cup \{0\}$	Same
Core constraints	$2^n - 1$	$ \mathcal{F} \leq 2^n - 1$
Recovers classical TSG	—	Yes, when $\mathcal{F} = 2^N \setminus \{\emptyset\}$

Definition 3 (Online TSG). An *Online Traveling Salesman Game* is a triple $(N, c, \mathcal{F})$ where N is the customer set, $c : 2^N \setminus \{\emptyset\} \rightarrow \mathbb{R}_+$ is the TSP characteristic cost defined on every nonempty subset, with core stability constraints imposed only for $S \in \mathcal{F}^p$, and $\mathcal{F} \subseteq 2^N \setminus \{\emptyset\}$ is the family from Definition 2.

An *allocation* is a vector $x \in \mathbb{R}^n$ with $\sum_i x_i = C(N)_{\text{online}}$, where $C(N)_{\text{online}}$ denotes the realized online tour cost. The *Temporal Core* is

$$\begin{aligned} \text{Core}(N, c, \mathcal{F}) := \{x \in \mathbb{R}^n : \\ \sum_i x_i = C(N)_{\text{online}}, \\ \sum_{i \in S} x_i \leq c(S) \ \forall S \in \mathcal{F}^p\}. \end{aligned}$$

When $C(N)_{\text{online}} = c^*(N)$, this definition coincides with the classical Potters et al. (1992) static TSG Core on the same constraint family. In general, the online overhead $C(N)_{\text{online}} - c^*(N) \geq 0$ raises the efficiency budget above the minimum grand-coalition cost; Table 2 summarizes the contrast, and the competitive ratio r (defined below) quantifies this overhead.

Characteristic cost versus realized cost. The characteristic function $c(S)$ is the time-independent optimal TSP tour on $S \cup \{0\}$, matching the classical Potters et al. (1992) definition. The realized online tour cost $C(N)_{\text{online}}$, by contrast, is the total travel distance of the tour actually executed by the online dispatch policy, which may exceed $c^*(N)$ because the online policy cannot anticipate future arrivals; in general $C(N)_{\text{online}} \geq c^*(N)$. We use $\sum_i x_i = C(N)_{\text{online}}$ as the efficiency budget because this is the cost the platform must actually allocate; the competitive ratio $r = C(N)_{\text{online}}/c^*(N)$ measures this overhead. The cost terms entering r^{**} (Theorem 9) are geometric through $c(\{i\})$, $c(N \setminus \{i\})$, and $c^*(N)$, while applicability and the feasible-complement index set $\{i : N \setminus \{i\} \in \mathcal{F}\}$ are determined by the realized arrival-service windows.

Illustrative example. Supplementary §S6 works through the Online TSG on the first five customers of the Solomon C101 benchmark (Solomon, 1987): the realized service timeline is fully sequential ($k = 1$), all five singletons are feasible while no complement coalition is, and the Temporal Nucleolus reduces to an equal-excess allocation across the singleton constraints, illustrating how temporal restriction discards 26 of the 31 static-TSG inequalities while preserving Core existence.

4 Cost Allocation: The Temporal Nucleolus

Even when the Core is nonempty it is typically a polytope, and one needs a selection rule. We adopt the nucleolus of [Schmeidler \(1969\)](#), adapted to the feasible family $\mathcal{F}$.

4.1 Definition

For an allocation x and $S \in \mathcal{F}^p$, the *excess* is $e(S, x) := \sum_{i \in S} x_i - c(S)$. Let $\theta(x)$ be the vector $(e(S, x))_{S \in \mathcal{F}^p}$ sorted in non-increasing order.

Definition 4 (Temporal Nucleolus). The Temporal Nucleolus of $(N, c, \mathcal{F})$ is

$$\text{TNu}(N, c, \mathcal{F}) := \operatorname{argmin}^{\text{lex}} \left\{ \theta(x) : \sum_i x_i = C(N)_{\text{online}} \right\},$$

where the minimization is lexicographic over the sorted excess vector.

4.2 Sequential LP computation

Following the standard technique of [Guajardo & Jörnsten \(2015\)](#), the Temporal Nucleolus is computed by solving at most $|\mathcal{F}^p|$ linear programs. At stage q the LP

$$\begin{aligned} \min_{x, \varepsilon} \quad & \varepsilon \\ \text{s.t.} \quad & \sum_i x_i = C(N)_{\text{online}}, \\ & \sum_{i \in S} x_i - c(S) \leq \varepsilon \quad \forall S \in \mathcal{F}^p \setminus \mathcal{F}_{< q}^p, \\ & \sum_{i \in S} x_i - c(S) = \varepsilon_{q(S)}^* \quad \forall S \in \mathcal{F}_{< q}^p, \end{aligned} \tag{1}$$

determines the q -th largest coordinate of θ . Here $\mathcal{F}_{< q}^p$ collects the coalitions whose excess was fixed at earlier stages with optimal value $\varepsilon_{q(S)}^*$. Finiteness of $\mathcal{F}^p$ guarantees termination in at most $|\mathcal{F}^p|$ stages; each LP has $n + 1$ variables and $|\mathcal{F}^p| + 1$ constraints.

Algorithmic scope. All Core-stability results of this paper depend only on the first-stage LP ε_1^* ; tie-breaking and primal-degeneracy refinements are needed only for the full Nucleolus point and are described in [Supplementary §S3](#).

The restriction from $2^N \setminus \{\emptyset\}$ to $\mathcal{F}$ materially reduces problem size whenever arrivals are spread in time. [Section 6](#) reports empirically $|\mathcal{F}|/(2^n - 1)$ as low as 0.061 for heavily sequential patterns.

4.3 Reduction to the static game at simultaneous arrivals

When all customers arrive simultaneously and the realized tour is additionally optimal ($C(N)_{\text{online}} = c^*(N)$), the Temporal Nucleolus reduces to the classical static Nucleolus of

Potters et al. (1992) on the unrestricted coalition family $2^N \setminus \{\emptyset\}$ (Supplementary Proposition 2, §S6). Numerical verification on 25 pattern-A instances of the main grid confirms coordinate-wise coincidence to within 10^{-9} . The Online TSG is therefore a proper generalization, with the static theory recovered as the $k = n$ extremal case where the peak waiting queue contains every customer at once.

In short, the Online TSG is a proper generalization: the static theory is the $k = n$ extremal case where the peak queue contains everybody at once.

5 Theoretical Results

We now develop the Core theory. Section 5.1 establishes conditions under which the Temporal Core is *empty*; Section 5.2 gives conditions under which it is guaranteed *nonempty*. Section 5.3 discusses the relationships between the parameters r , r^{**} , k , and the queue dynamics.

5.1 Conditions under which the Core vanishes

Remark 5 (Necessary super-additivity failure). A trivial necessary condition for Temporal Core emptiness comes from super-additivity failure of the singleton constraints: if $C(N)_{\text{online}} > \sum_{i \in N} c(\{i\})$, equivalently $r > r^* := \sum_i c(\{i\})/c^*(N)$, then no allocation can simultaneously satisfy the efficiency equality $\sum_i x_i = C(N)_{\text{online}}$ and the singleton constraints $x_i \leq c(\{i\})$, so the Temporal Core is empty. This bound is rarely tight in practice (r^* is typically in the range 3–4 while the competitive ratio of reasonable online policies is close to 1); we sharpen it substantially by exploiting complement coalitions $N \setminus \{i\}$. The feasibility characterization of complement coalitions and a structural consequence under fully sequential arrivals are recorded as Supplementary Lemmas 4 and 5 (§S8); we cite them where used.

Theorem 9 (Sufficient condition for empty Temporal Core via complement coalitions). *Assume that $N \setminus \{i\} \in \mathcal{F}$ for some $i \in N$. Let*

$$\begin{aligned} \delta_i &:= c(\{i\}) + c(N \setminus \{i\}) - c^*(N), \\ r^{**} &:= 1 + \min_{i: N \setminus \{i\} \in \mathcal{F}} \delta_i / c^*(N). \end{aligned}$$

If $r = C(N)_{\text{online}}/c^(N) > r^{**}$, then the Temporal Core is empty.*

Proof. Let $i^* \in \arg \min_{i: N \setminus \{i\} \in \mathcal{F}} \delta_i$. Both $\{i^*\} \in \mathcal{F}$ (singletons are always feasible) and $N \setminus \{i^*\} \in \mathcal{F}$. Suppose, for contradiction, that x lies in the Temporal Core. The two core inequalities read

$$x_{i^*} \leq c(\{i^*\}), \tag{2}$$

$$\sum_{j \neq i^*} x_j \leq c(N \setminus \{i^*\}). \tag{3}$$

Adding (2) and (3) gives $\sum_{j \in N} x_j \leq c(\{i^*\}) + c(N \setminus \{i^*\}) = c^*(N) + \delta_{i^*}$. But the efficiency constraint forces $\sum_j x_j = C(N)_{\text{online}} = r \cdot c^*(N)$. Hence $r \leq 1 + \delta_{i^*}/c^*(N) = r^{**}$, contradicting the hypothesis $r > r^{**}$. $\square$

Because $\min_i \delta_i$ is typically small ($\delta_i = 0$ corresponds to $\{i\}, N \setminus \{i\}$ partitioning the grand tour with no overhead), r^{**} is close to 1 and thus an operationally informative threshold. We stress that Theorem 9 is tight only *within the single-complement mechanism*: even when no single complement $N \setminus \{i\}$ alone violates the Core, a balanced collection of complements can; this is captured by Proposition 10 below. Intermediate-size coalitions S with $|S| \in \{2, \dots, n-2\}$ can also independently violate the Core inequality $\sum_{i \in S} x_i \leq c(S)$; Section 6 exhibits observed cases of both mechanisms (Remark 12 and §6.3).

Proposition 10 (Balanced-complement threshold). *Let $n \geq 2$ and suppose every complement coalition is feasible, i.e., $N \setminus \{i\} \in \mathcal{F}$ for all $i \in N$. Define*

$$r^{***} := \frac{1}{n-1} \cdot \frac{\sum_{i \in N} c(N \setminus \{i\})}{c^*(N)}.$$

If $r = C(N)_{\text{online}}/c^(N) > r^{***}$, then the Temporal Core is empty.*

Proof. Suppose $x \in \text{Core}(N, c, \mathcal{F})$. For each $i \in N$, the Core constraint on $N \setminus \{i\}$ reads $\sum_{j \neq i} x_j \leq c(N \setminus \{i\})$. Summing over $i \in N$ and exchanging the order of summation gives

$$(n-1) \sum_{j \in N} x_j = \sum_{i \in N} \sum_{j \neq i} x_j \leq \sum_{i \in N} c(N \setminus \{i\}).$$

The efficiency equality $\sum_j x_j = C(N)_{\text{online}}$ yields $(n-1) C(N)_{\text{online}} \leq \sum_i c(N \setminus \{i\})$. Dividing by $(n-1) c^*(N)$ produces $r \leq r^{***}$, contradicting $r > r^{***}$. $\square$

The collection $\{N \setminus \{i\}\}_{i \in N}$ is balanced with weights $\lambda_i = 1/(n-1)$ (each player appears in exactly $n-1$ complements), so Proposition 10 is the Bondareva–Shapley criterion (Shapley, 1971) specialised to this collection. Theorem 9 and Proposition 10 are incomparable in general: Theorem 9 depends on the minimum δ_i and fires when a single complement is cheap, whereas Proposition 10 averages the complement costs and fires when the balanced collection as a whole is cheap. Section 6 exhibits instances where $r > r^{***}$ but $r \leq r^{**}$, so the Core is certified empty by Proposition 10 alone.

Corollary 11 (Fully sequential arrivals disarm Theorem 9). *If arrivals are fully sequential in the sense of Supplementary Lemma 5, then Theorem 9 is vacuous: no $N \setminus \{i\}$ is feasible, so its hypothesis cannot hold.*

Proof. Direct from Supplementary Lemma 5 and the hypothesis of Theorem 9. The same argument rules out Proposition 10’s hypothesis, which additionally requires every $N \setminus \{i\}$ to be feasible; the balanced-complement mechanism is likewise blocked under fully sequential arrivals. $\square$

Remark 12 (Near-complement mechanism). Beyond Theorem 9 and Proposition 10, we observe main-grid instances (10 of 175 under the NN policy) where the Temporal Core is empty, both thresholds are vacuous, and the binding constraints in the Core LP mix size- $(n-1)$ and size- $(n-2)$ coalitions simultaneously. Proposition 13 below formalises this mechanism as a Bondareva–Shapley sufficient condition on the mixed collection $B_{n-1} \cup B_{n-2}$; it fires on 9 of the 10 observed cases (Section 6.3, Table 4; Supplementary Materials §S4).

Proposition 13 (Balanced-near-complement threshold). *Let $n \geq 3$, $B_{n-1} = \{N \setminus \{i\} : i \in N\}$, and $B_{n-2} = \{N \setminus \{i, j\} : i \neq j \in N\}$. Assume $\mathcal{F} \supseteq B_{n-1} \cup B_{n-2}$, i.e. every size- $(n-1)$ and size- $(n-2)$ coalition is feasible. Let $\lambda = (\lambda_S)_{S \in B_{n-1} \cup B_{n-2}}$ be a vector of non-negative weights satisfying the balancing condition*

$$\sum_{S \ni i} \lambda_S = 1 \quad \text{for every } i \in N,$$

where the sum is taken over all $S \in B_{n-1} \cup B_{n-2}$ containing i . Define

$$r^{(\diamond)}(\lambda) := \frac{\sum_{S \in B_{n-1} \cup B_{n-2}} \lambda_S c(S)}{c^*(N)}.$$

If the realized competitive ratio $r = C(N)_{\text{online}}/c^*(N)$ satisfies $r > r^{(\diamond)}(\lambda)$ for some such λ , then $\text{Core}(N, c, \mathcal{F}) = \emptyset$.

Proof. Suppose for contradiction that $x \in \text{Core}(N, c, \mathcal{F})$. The feasibility hypothesis $\mathcal{F} \supseteq B_{n-1} \cup B_{n-2}$ ensures that for every $S \in B_{n-1} \cup B_{n-2}$ the Temporal Core stability inequality

$$\sum_{i \in S} x_i \leq c(S) \tag{4}$$

is imposed. Multiplying (4) by the non-negative weight λ_S and summing over $S \in B_{n-1} \cup B_{n-2}$ yields

$$\sum_{S \in B_{n-1} \cup B_{n-2}} \lambda_S \sum_{i \in S} x_i \leq \sum_{S \in B_{n-1} \cup B_{n-2}} \lambda_S c(S).$$

The left-hand side is a finite double sum; interchanging the order of summation gives

$$\sum_S \lambda_S \sum_{i \in S} x_i = \sum_{i \in N} x_i \sum_{S \ni i} \lambda_S = \sum_{i \in N} x_i,$$

where the second equality uses the balancing condition $\sum_{S \ni i} \lambda_S = 1$ for every $i \in N$. The Temporal Core efficiency equality (Definition 3) gives $\sum_{i \in N} x_i = C(N)_{\text{online}} = r c^*(N)$. Combining,

$$r c^*(N) \leq \sum_{S \in B_{n-1} \cup B_{n-2}} \lambda_S c(S) = r^{(\diamond)}(\lambda) \cdot c^*(N),$$

i.e. $r \leq r^{(\diamond)}(\lambda)$, contradicting the hypothesis $r > r^{(\diamond)}(\lambda)$. Therefore $\text{Core}(N, c, \mathcal{F}) = \emptyset$. $\square$

Setting $\lambda_S = 1/(n-1)$ on B_{n-1} and $\lambda_S = 0$ on B_{n-2} recovers Proposition 10 on the subdomain

where Proposition 13’s feasibility hypothesis $\mathcal{F} \supseteq B_{n-1} \cup B_{n-2}$ holds; on this subdomain Proposition 13 is a refinement, optimizing λ over the larger collection. Note that the two propositions have different applicability domains: Proposition 10 requires only $\mathcal{F} \supseteq B_{n-1}$, while Proposition 13 additionally requires $\mathcal{F} \supseteq B_{n-2}$. The sharpest threshold $r_{\min}^{(\diamond)}$ is the optimum of a linear program with $n + \binom{n}{2}$ variables and n equality constraints; Section 6.3 reports the empirical fire-count (9 of 10).

Proposition 14 (General partition-pair threshold). *For any partition pair $(S, N \setminus S)$ with $S \in \mathcal{F}$, $N \setminus S \in \mathcal{F}$, and $1 \leq |S| \leq n - 1$, define*

$$r_S^{(P)} := \frac{c(S) + c(N \setminus S)}{c^*(N)},$$

and let $\mathcal{P}(\mathcal{F}) := \{(S, N \setminus S) : S, N \setminus S \in \mathcal{F}\}$ collect the feasible partition pairs. Set $r_{\min}^{(P)} := \min_{(S, N \setminus S) \in \mathcal{P}(\mathcal{F})} r_S^{(P)}$. If the realized competitive ratio r satisfies $r > r_{\min}^{(P)}$, then $\text{Core}(N, c, \mathcal{F}) = \emptyset$.

Proof. The collection $\{S, N \setminus S\}$ with weights $\lambda_S = \lambda_{N \setminus S} = 1$ is balanced: every $i \in N$ lies in exactly one of the two coalitions, so $\sum_{T \ni i} \lambda_T = 1$. Suppose for contradiction $x \in \text{Core}(N, c, \mathcal{F})$. Multiplying the stability inequalities $\sum_{i \in T} x_i \leq c(T)$ for $T \in \{S, N \setminus S\}$ by $\lambda_T = 1$ and summing, then exchanging the order of summation using the balancing identity, yields $\sum_{i \in N} x_i \leq c(S) + c(N \setminus S)$. Combined with the efficiency equality $\sum_i x_i = C(N)_{\text{online}} = r c^*(N)$, this gives $r c^*(N) \leq c(S) + c(N \setminus S)$, i.e. $r \leq r_S^{(P)}$, contradicting $r > r_{\min}^{(P)}$. $\square$

Relationship to earlier thresholds. Theorem 9 is the $|S| = 1$ specialization of Proposition 14: with $S = \{i\}$, $r_{\{i\}}^{(P)} = 1 + \delta_i / c^*(N)$ recovers r^{**} , and the feasibility hypothesis $S, N \setminus S \in \mathcal{F}$ reduces to $N \setminus \{i\} \in \mathcal{F}$. The novelty of Proposition 14 is that $r_{\min}^{(P)}$ remains *defined* when no *singleton* complement is feasible, provided some *non-singleton* partition pair has both halves in $\mathcal{F}$. This is precisely the regime $k < n - 1$: although every $N \setminus \{i\}$ is infeasible (Corollary 18), a partition $(S, N \setminus S)$ with $|S|, |N \setminus S| \in \{2, \dots, n - 2\}$ can still have both halves admissible. Proposition 14 is therefore the first Bondareva–Shapley threshold surviving the structural obstruction of Corollary 18; the empirical content of Observation 15 below is exactly that this regime is non-empty. Operationally, Proposition 14 is most naturally read as a *certificate* rather than a predictive bound: given a feasible partition pair, it provides analytic verification of Core emptiness, but locating the partition pair $(S, N \setminus S)$ that minimizes $r_S^{(P)}$ in general requires either an enumeration over feasible partition pairs of $\mathcal{F}$ or extraction from the dual support of the first-stage Core LP (see Supplementary §S9).

Observation 15 (Intermediate-coalition mechanism, certified by Proposition 14). In 9 of the 175 main-grid instances under the NN policy, the Temporal Core is empty despite (i) the hypothesis of Theorem 9 being vacuous ($r \leq r^{**}$, or no singleton complement feasible) and (ii) the hypothesis of Proposition 10 being vacuous. All 9 satisfy $k < n - 1$ and arise in patterns B_{medium} (7 cases) and E (2 cases) with realized competitive ratio $r \in [1.33, 1.61]$. The binding constraints in the Core LP concentrate on intermediate-size coalitions ($2 \leq |S| \leq n - 3$) rather than on complements or

near-complements. The first-stage Core-LP dual of each of the 9 cases is supported on exactly one partition pair $(S, N \setminus S)$ with $|S|, |N \setminus S| \in \{2, \dots, n-2\}$; the corresponding partition-pair threshold $r_S^{(P)}$ falls strictly below the realized r in every case (margins 0.008–0.152), so Proposition 14 certifies all 9 cases analytically. Detailed per-case partition pairs appear in Supplementary §S9.

Theorem 16 (Asymptotic tightening of r^{**}). *Assume customer coordinates $x_1, \dots, x_n$ are drawn i.i.d. from the uniform distribution on $[0, L]^2$, with the depot placed at the origin $(0, 0)$. Suppose the arrival pattern is such that $N \setminus \{i\} \in \mathcal{F}$ for at least one index i . Then*

$$r^{**} = 1 + \frac{\min_{i: N \setminus \{i\} \in \mathcal{F}} \delta_i}{c^*(N)} \xrightarrow{\text{a.s.}} 1 \quad \text{as } n \rightarrow \infty,$$

with rate $r^{**} - 1 = O(n^{-1/2})$ almost surely, uniformly over the subclass of arrival patterns for which $\{i \in N : N \setminus \{i\} \in \mathcal{F}\}$ is non-empty.

Proof. Fix any index i^0 with $N \setminus \{i^0\} \in \mathcal{F}$. We bound δ_{i^0} first, then scale by $c^*(N)$.

Step 1: insertion and monotonicity bounds. Inserting the detour depot $\rightarrow x_{i^0} \rightarrow$ depot into any tour of $N \setminus \{i^0\}$ shows

$$c^*(N) \leq c(N \setminus \{i^0\}) + c(\{i^0\}), \quad (5)$$

whence $c^*(N) - c(N \setminus \{i^0\}) \leq c(\{i^0\}) = 2d(0, x_{i^0})$. Conversely, removing x_{i^0} from the optimal tour on N and short-cutting its two tour neighbors yields a valid Hamiltonian cycle on $N \setminus \{i^0\} \cup \{0\}$ of no greater length, by the triangle inequality (Steele, 1997, §2.1); hence $c(N \setminus \{i^0\}) \leq c^*(N)$ and $c^*(N) - c(N \setminus \{i^0\}) \geq 0$. Combining:

$$\begin{aligned} \delta_{i^0} &:= c(\{i^0\}) + c(N \setminus \{i^0\}) - c^*(N) \geq 0 \quad [\text{insertion bound, eq. (5)}], \\ \delta_{i^0} &\leq c(\{i^0\}) \quad [\text{monotonicity, since } c^*(N) \geq c(N \setminus \{i^0\})], \\ c(\{i^0\}) &= 2d(0, x_{i^0}). \end{aligned}$$

Since $x_{i^0} \in [0, L]^2$, $d(0, x_{i^0}) \leq L\sqrt{2}$ and thus $\delta_{i^0} \leq 2L\sqrt{2} = O(1)$, independent of the arrival pattern.

Step 2: BHH lower bound on $c^(N)$.* By the Beardwood–Halton–Hammersley theorem (Beardwood et al., 1959; Steele, 1981), the length of the optimal Hamiltonian cycle through n i.i.d. uniform points in $[0, L]^2$ satisfies

$$\frac{c^*(N)}{\sqrt{n}} \xrightarrow{\text{a.s.}} \beta_{\text{BHH}} L \quad \text{as } n \rightarrow \infty,$$

for a universal constant $\beta_{\text{BHH}} > 0$. Adding a fixed depot shifts the length by $O(1)$, which is absorbed into the $o(\sqrt{n})$ term; hence $c^*(N) = \Theta(\sqrt{n})$ a.s.

Step 3: ratio. Combining Steps 1–2,

$$\begin{aligned}
r^{**} - 1 &= \frac{\min_{i: N \setminus \{i\} \in \mathcal{F}} \delta_i}{c^*(N)} \leq \frac{\delta_{i_0}}{c^*(N)} \\
&\leq \frac{2L\sqrt{2}}{\beta_{\text{BHH}} L \sqrt{n} (1 + o(1))} \\
&= O(n^{-1/2}) \quad \text{a.s.}
\end{aligned}$$

The numerator bound from Step 1 uses only the fixed coordinates $x_{i_0} \in [0, L]^2$ and not the arrival pattern, so the rate holds uniformly across the subclass of arrival patterns for which $\{i \in N : N \setminus \{i\} \in \mathcal{F}\}$ is non-empty (patterns outside this subclass have r^{**} undefined and the claim is vacuous). Non-negativity $r^{**} - 1 \geq 0$ from Step 1 completes the claim.

Relation to the N2 convention. The proof is stated for an arbitrary fixed coordinate square $[0, L]^2$, yielding $\delta_{i_0} = O(L)$ (Step 1) and $c^*(N) = \Theta(L\sqrt{n})$ (Step 2). The parameter L therefore cancels exactly in the Step 3 ratio, so the $O(n^{-1/2})$ rate is independent of the choice of L as a function of n . Under the N2 convention $L = \sqrt{n}$ adopted in Section 6.1, the numerator becomes $O(\sqrt{n})$ and the denominator $\Theta(n)$, whose ratio is the same $O(n^{-1/2})$. The theorem applies verbatim to the experimental setting. $\square$

The i.i.d. uniform hypothesis is required for the BHH $\sqrt{n}$ scaling of $c^*(N)$ in Step 2; for clustered or other non-uniform absolutely continuous distributions, an analogous $O(n^{-1/2})$ rate holds by the general limiting-functional framework of Steele (1997), with the hidden constant then depending on the density.

Remark 17 (Faster decay when a depot-close complement is feasible). Let $i^* \in \arg \min_i d(0, x_i)$ denote the customer closest to the depot. Under i.i.d. uniform sampling on $[0, L]^2$, the minimum of n points' distances to the origin satisfies $\min_i d(0, x_i) = \Theta_P(n^{-1/2})$ —i.e., bounded above and below at rate $n^{-1/2}$ in probability; $\sqrt{n} \min_i d(0, x_i)$ converges in distribution to a non-degenerate Rayleigh-type limit, so an almost-sure Θ bound does not hold, and the radial density of uniform points near the depot satisfies $p(r) \propto r$ as $r \rightarrow 0$ (whether the depot sits inside the square or at its corner, yielding the same $n^{-1/2}$ rate up to a universal constant depending on the sector angle swept by the square at the depot— 2π for an interior depot, $\pi/2$ for a corner depot, with intermediate values for boundary (non-corner) placements; this constant ratio therefore lies in $[1, 4]$ between the corner and interior cases but does not change the $n^{-1/2}$ in-probability rate). If, in addition, $N \setminus \{i^*\} \in \mathcal{F}$ —a condition automatically met when all customers share a waiting window, and more broadly whenever the arrival pattern admits the depot-closest customer's complement—Step 1 of the proof applied to i^* yields $\delta_{i^*} \leq 2d(0, x_{i^*}) = O_P(n^{-1/2})$, hence

$$r^{**} - 1 = \frac{\min_i \delta_i}{c^*(N)} \leq \frac{\delta_{i^*}}{c^*(N)} = O_P(n^{-1}),$$

a faster rate than the generic $O(n^{-1/2})$ of Theorem 16; only the hidden constant, and not the

rate, depends on depot position within $[0, L]^2$. An almost-sure version with an extra log factor, $r^{**} - 1 = O(\log n/n)$, follows from a standard Borel–Cantelli argument, which we omit; the O_P bound suffices for the operational message.

Operationally, the faster $O(n^{-1})$ rate applies primarily to arrival patterns close to the simultaneous regime, where Theorem 9’s hypothesis is easily met. In sparser regimes (sequential patterns, large queue gaps) r^{**} is not defined in the first place—no $N \setminus \{i\}$ need be feasible—and the Remark is vacuous; Corollary 18’s structural obstruction governs those regimes.

5.2 Conditions under which the Core survives

A trivial inheritance result holds in the special case of exact online dispatch: when $C(N)_{\text{online}} = c^*(N)$ every static-TSG Core allocation remains in the Temporal Core, since efficiency matches and every $\mathcal{F}^P$ -constraint is already implied by the static Core (Supplementary Proposition 3, §S7). Operationally this is informative only in the simultaneous-arrival regime where it coincides with the static Core itself; the structurally informative survival statement is the obstruction at $k < n - 1$.

Corollary 18 (Structural obstruction of the complement mechanisms). *If the peak waiting queue satisfies $k = \max_t |U_t| < n - 1$, then no coalition of the form $N \setminus \{i\}$ is feasible, so Theorem 9, Proposition 10, and Proposition 13 are each vacuous: none of their feasibility hypotheses can hold. The only remaining a priori mechanism for emptying the Temporal Core under $k < n - 1$ is Remark 5, which is dominated by r^* and rarely binds in practice. Intermediate-coalition violations remain logically possible; their empirical frequency is characterized in Section 6.*

Proof. If $N \setminus \{i\} \in \mathcal{F}$ for some i , then $|U_t| \geq n - 1$ at some t , contradicting $k < n - 1$. Theorem 9’s hypothesis requires one feasible complement, and Propositions 10 and 13 require $B_{n-1} \subseteq \mathcal{F}$ (strictly stronger); each is therefore violated. The final claim follows from Theorem 9 and Remark 5. $\square$

5.3 Discussion

Parameters interact as follows.

- The *tight* threshold r^{**} is sensitive only to the cheapest complement coalition: a single pair $(i, N \setminus \{i\}) \in \mathcal{F}$ with small δ_i is enough to expose the Core.
- The structural parameter k bounds queue depth. Since $N \setminus \{i\}$ -feasibility requires $|U_t| \geq n - 1$, the inequality $k < n - 1$ is exactly the combinatorial negation of Theorem 9’s hypothesis.
- Arrival patterns interpolate between two extremes: simultaneous ($k = n$, $\mathcal{F} = 2^N \setminus \{\emptyset\}$, classical TSG) and fully sequential ($k \leq 1$ for sequential arrival regimes, Theorem 9 vacuous).

In queueing-theoretic terms, $k \geq n - 1$ with positive probability activates the complement-coalition mechanism; otherwise Corollary 18 blocks it, though intermediate-coalition and near-complement violations (Remark 12) remain logically possible.

6 Computational Experiments

We report a computational study whose purpose is to (i) verify that the theoretical conditions fire exactly as predicted, (ii) quantify how much sharper r^{**} is than r^* , and (iii) characterize the empirical frequency of Core existence across realistic arrival regimes.

6.1 Design

Dimensionless framework. Instances follow the standard dimensionless normalization of [Steele \(1997\)](#): customers are sampled uniformly in $[0, L]^2$ with $L = \sqrt{n}$, fixing the customer density at one per unit area. Vehicle speed is normalized to one (unit distance per unit time), and the depot is at the origin. The system is governed by a single dimensionless load parameter

$$\rho = \frac{L/v}{\tau} = \frac{\text{characteristic travel time}}{\text{mean inter-arrival interval}},$$

where τ is the mean inter-arrival interval of a given pattern (with vehicle speed normalized to $v = 1$, $\rho = L/\tau$). All theoretical bounds and empirical statistics depend on (n, ρ) alone; the joint rescaling $(L, v, \tau) \rightarrow (\alpha L, \alpha v, \alpha \tau)$ for any $\alpha > 0$ leaves all observables invariant. We verify this invariance empirically to machine precision in Supplementary Materials §S2.

Arrival patterns. Seven arrival patterns span the ρ -spectrum:

- A ($\rho = \infty$): all customers arrive simultaneously at $t = 0$.
- B_{heavy} ($\rho = 4$), B_{medium} ($\rho = 2$), B_{light} ($\rho = 0.5$): sequential arrivals at uniform interval $\tau = \beta_{BHH}/\rho$, where $\beta_{BHH} \approx 0.7124$ is the Beardwood–Halton–Hammersley constant.
- C ($\rho = 2$): clustered arrivals in two angular cohorts, interleaved in space.
- D ($\rho = 1$): reverse-angle zig-zag order.
- E ($\rho = 1$): uniformly random arrivals in $[0, n\tau]$.

The seven patterns form a $(\rho, \text{structure})$ factorial rather than a ρ -only sweep: the load axis is traced by A and the three uniform-sequential B-rates ($\rho \in \{0.5, 1, 2, 4, \infty\}$); at the saturation-borderline $\rho = 2$ a uniform-sequential pattern (B_{medium}) and a clustered pattern (C) are paired, and at $\rho = 1$ a reverse-angle pattern (D) and a random pattern (E) are paired, isolating arrival geometry from arrival rate at fixed ρ . Patterns B_{medium} and C share $\rho = 2$ but differ in arrival structure; [Section 6.4](#) shows that this structural difference alone suffices to place them in qualitatively distinct regimes of Core stability, illustrating that the load parameter ρ is necessary but not sufficient to characterize dynamics.

Table 3: Experimental design: 7 patterns $\times$ 5 sizes $\times$ 5 seeds = 175 instances per dispatch policy; 3 policies yield 525 policy-instance pairs.

Factor	Levels
n (coalition size)	5, 7, 10, 12, 15
Arrival pattern	A ($\rho = \infty$), B _{heavy} ($\rho = 4$), B _{medium} ($\rho = 2$), B _{light} ($\rho = 0.5$), C ($\rho = 2$, clustered), D ($\rho = 1$, reverse), E ($\rho = 1$, random)
Seed	{7, 42, 99, 123, 256}
Dispatch policy	Nearest-neighbor (NN), cheapest-insertion (CI), batch re-optimization (BR)

Table 4: Five-mechanism contingency on the 175-instance NN grid. The three complement-based analytic thresholds (Theorem 9, Proposition 10, Proposition 13) jointly certify 57 of the 67 observed empty Cores, and the general partition-pair threshold (Proposition 14) certifies the remaining 9 intermediate-coalition cases of Observation 15 that operate at $k < n - 1$ (66 of 67 jointly). One near-complement case eludes all four (Remark 12 residual). No false positives arise under any policy. Row labels denote the best-discriminating mechanism per instance, not the only threshold that fires. Standalone fire counts overlap: under NN dispatch, Proposition 10’s bound $r > r^{***}$ is satisfied in 57 of the 175 instances (all 37 Theorem 9 fires plus 20 further cases, of which 11 lack any size- $(n - 2)$ binding and are labeled “balanced-complement” (best label) and 9 also exhibit size- $(n - 2)$ binding and are labeled “balanced-near-complement” (best label)). Proposition 13 contains the Proposition 10 weighting on the common applicability domain (setting $\lambda_S = 1/(n - 1)$ on $\mathcal{B}_{n-1}$, zero on $\mathcal{B}_{n-2}$ recovers Proposition 10; see Section 5.1), specifically including the 9 “balanced-near-complement” (best label) cases listed above.

Classification	Core empty (obs.)	Core nonempty (obs.)	Total
Single-complement (Theorem 9 fires)	37	0	37
Balanced-complement (Proposition 10, best label)	11	0	11
Balanced-near-complement (Proposition 13, best label)	9	0	9
Near-complement residual (Remark 12; no threshold fires)	1	0	1
Intermediate (Observation 15)	9	0	9
No mechanism predicts emptiness	0	108	108
Total	67	108	175

Computational setup. The online tour is computed by the nearest-neighbor heuristic; Section 6.6 compares this baseline with cheapest-insertion and batch re-optimization policies. The static TSP $c^*(S)$ for each S is computed exactly by dynamic programming (Held–Karp) for $|S| \leq 15$ and by the LKH heuristic (Helsgaun, 2000) otherwise. The Temporal Nucleolus is computed by the sequential LP (1) implemented with PuLP and the CBC solver.

Table 3 summarizes the design: 7 patterns $\times$ 5 sizes $\times$ 5 seeds = 175 instances per dispatch policy, and 3 policies yield 525 policy-instance pairs in the main study, plus an additional scale-up grid at $n \in \{20, 30, 50\}$ reported in Section 6.7.

6.2 Verification of Theorem 9

Table 4 is the five-mechanism contingency of predicted versus observed Core status under NN dispatch on the full 175-instance main grid.

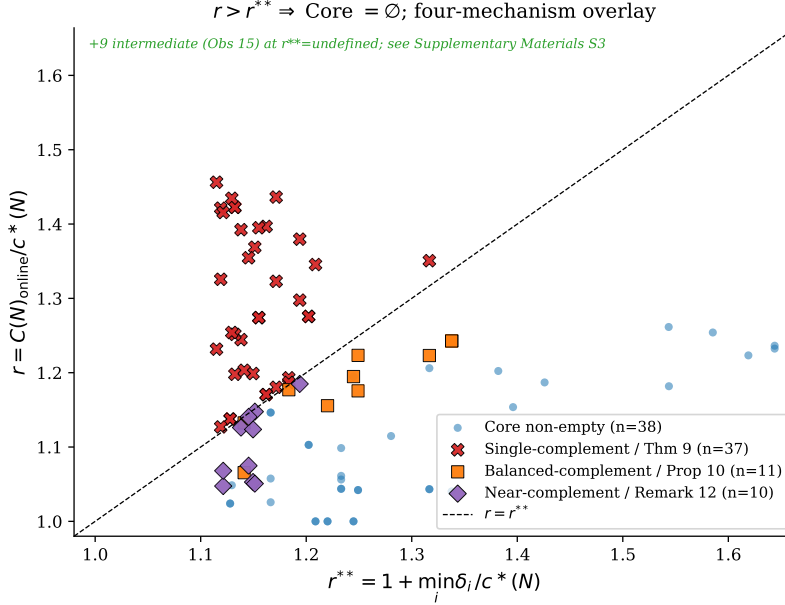


Figure 1: Empirical test of Theorem 9 and Proposition 10. The dashed line is $r = r^{**}$; 37 instances strictly above the diagonal have an empty Core (red $\times$, single-complement mechanism). Of the 10 additional empty-Core instances plotted here as near-complement (purple diamond), 9 are certified by Proposition 13 (balanced-near-complement threshold on $B_{n-1} \cup B_{n-2}$) and 1 remains covered only by the first-stage Core LP (Remark 12 residual); the 11 balanced-complement cases covered by Proposition 10 (orange square) sit at or below the diagonal. The nine intermediate-coalition cases of Observation 15 have r^{**} undefined and are not plotted. See Section 6.3 for the full five-mechanism decomposition. The legend’s “Core non-empty (n=38)” counts only the 96 instances on which r^{**} is defined; pooled with the 9 intermediate-coalition cases and the remaining nonempty instances on which r^{**} is undefined, this is consistent with the overall 108 nonempty Cores reported in Table 4.

Every instance classified by any of the five mechanisms as “empty” is indeed empty (no false positives across 525 policy-instance pairs). The three complement-based mechanisms cover 57 of 67 NN empties; Proposition 14 certifies the 9 intermediate cases of Observation 15 (Supplementary §S9), so the four thresholds jointly cover 66 of 67. Only the 1 near-complement residual (Remark 12, $r \leq r_{\min}^{(\circ)}$ by $\approx 2 \times 10^{-3}$) is uncovered analytically. Figure 1 plots r versus r^{**} : instances strictly above the diagonal ($r > r^{**}$) are the single-complement mechanism, while balanced-complement, balanced-near-complement, near-complement-residual, and intermediate cases lie at or below the diagonal and are covered by their respective certificates. The sharpness of r^{**} and r^{***} relative to r^* is illustrated in Supplementary Figure S2 (§S10).

6.3 Verification of Corollary 18 and observed intermediate-coalition mechanism

Analytic sharpness of Corollary 18. The analytic content of Corollary 18 is that all three complement-based mechanisms—single (Theorem 9), balanced (Proposition 10), and balanced-near-complement (Proposition 13)—are blocked whenever the peak queue satisfies $k < n - 1$,

because each hypothesis requires at least one complement $N \setminus \{i\}$ to be feasible, and feasibility forces $|U_t| = n - 1$ at some t . Empirically this holds without exception in the 175-instance grid: no NN (or CI, BR) instance with $k < n - 1$ satisfies $r > r^{**}$, $r > r^{***}$, or $r > r_{\min}^{(\diamond)}$ (false-positive count is zero for all three thresholds across all 525 policy-instance pairs).

Peak-queue strata. Partitioning the 175 main-grid instances (NN) by peak-queue regime yields three strata (Figure 2):

- $k < n - 1$: 79 instances, 70 Core nonempty (88.6%).
- $k = n - 1$: 16 instances, 16 Core nonempty (100%).
- $k = n$: 80 instances, 22 Core nonempty (27.5%).

A per-pattern breakdown across $n \in \{5, 7, 10, 12, 15\}$ appears in Supplementary Materials §S1 (Supplementary Figure S1).

Nine empty Cores at $k < n - 1$: intermediate-coalition mechanism. The 9 empty-Core instances in the $k < n - 1$ regime are all certified by the intermediate-coalition mechanism of Observation 15. All 9 lie in B_{medium} (7 cases) or E (2 cases), have k between $n - 5$ and $n - 2$, and realize $r \in [1.33, 1.61]$. Since both Theorem 9 and Proposition 10 are vacuous in the $k < n - 1$ regime, the 9 cases constitute direct evidence of a third mechanism operating outside the complement-coalition framework. Table 5 summarises these instances; full binding-size distributions are deferred to Supplementary Materials §S3. For the 9 cases the partition pairs were obtained from the dual support of the first-stage Core LP (Supplementary §S9); a predictive analytic characterization of which arrival-service realizations admit such a partition pair, derived from queueing-theoretic primitives rather than from after-the-fact LP duals, is open (Section 7, Limitation (iv)).

Table 5: Nine main-grid instances with $k < n - 1$ and empty Core (intermediate-coalition mechanism, Observation 15). All arise under NN dispatch in patterns B_{medium} or E. The “dominant size” column reports the coalition size at which the largest number of constraints are tight at the first-stage LP optimum.

n	pattern	seed	k	$n - 1$	r	ε^*	dominant size
7	B_{medium}	256	5	6	1.436	0.028	4
10	B_{medium}	123	8	9	1.423	0.192	5 (tied with 4)
10	B_{medium}	256	8	9	1.358	0.217	7 (tied with 6)
10	E	42	8	9	1.495	0.062	3
12	B_{medium}	99	10	11	1.395	0.744	6 (tied with 7)
12	B_{medium}	123	10	11	1.487	0.767	6
12	B_{medium}	256	10	11	1.423	0.941	7 (tied with 8)
12	E	42	9	11	1.610	0.289	4 (tied with 5, 6)
15	B_{medium}	256	12	14	1.326	0.185	9 (tied with 10)

Five-way decomposition of 67 empty Cores (NN). The 67 empty Cores decompose into five mechanisms:

- Single-complement (Theorem 9): **37** instances, all at $k = n$.
- Balanced-complement (Proposition 10): **11** instances, all at $k = n$.
- Balanced-near-complement (Proposition 13): **9** instances, all at $k = n$.
- Near-complement residual (Remark 12, no threshold fires): **1** instance ($n = 15$, Pattern A, seed 42; $r = 1.0475$, $r_{\min}^{(\diamond)} = 1.0495$), at $k = n$.
- Intermediate (Observation 15): **9** instances, all at $k < n - 1$.

Two regimes emerge: the saturated regime $k = n$ concentrates the four complement-based mechanisms (single + balanced + balanced-near-complement + near-complement-residual: 58 of 58 empty cases at $k = n$), while the shallow-queue regime $k < n - 1$ is governed exclusively by the intermediate mechanism. These regimes are empirically disjoint in our grid.

Practical implication. Corollary 18 guarantees that keeping $k < n - 1$ eliminates the complement-coalition mechanisms. In this regime, however, the intermediate mechanism of Observation 15 remains active and can empty the Core when the realized competitive ratio is large. Crucially, all 9 intermediate cases arise under nearest-neighbor dispatch: under cheapest-insertion and batch re-optimization, no intermediate-coalition empty Cores are observed in our grid (Section 6.6). The intermediate mechanism thus appears *policy-sensitive* and can be mitigated operationally, though an analytic threshold is open.

6.4 Pattern-level analysis

Table 6 reports the salient statistics per arrival pattern, pooled over n and seeds. Figure 3 visualizes the relative size of $\mathcal{F}$.

Three patterns stand out. Simultaneous arrivals (A, $\rho = \infty$) yield the lowest Core-existence rate (36%), consistent with $\mathcal{F} = 2^N \setminus \{\emptyset\}$ maximizing the number of Core constraints. Light-load sequential B_{light} ($\rho = 0.5$) leaves $|\mathcal{F}|/(2^n - 1)$ at only 0.077 and renders Theorem 9 inapplicable in all 25 instances. Mid-dispersal patterns D and E also keep $|\mathcal{F}|$ small (0.196 and 0.273 respectively), with Theorem 9 applicable in only 3/25 and 5/25 instances and firing in 0/25 in both. The clustered pattern C and the heavy-load B_{heavy} have $\rho \geq 2$ and near-saturated feasibility ($|\mathcal{F}|/(2^n - 1) = 0.985$ each), placing them in the regime where Theorem 9 applies to nearly every instance and fires in 10/25 (C) and 15/25 (B_{heavy}).

A secondary observation: the mean competitive ratio $\bar{r}$ rises sharply in the heavily sequential regimes (B_{light} : 1.69, E: 1.34, D: 1.33) because the online tour takes detours to reach customers as they arrive while $c^*(N)$ reflects the offline-optimal route. In patterns B_{light} and D the queue stays shallow throughout, and Corollary 18 blocks every complement-based mechanism; the Core remains

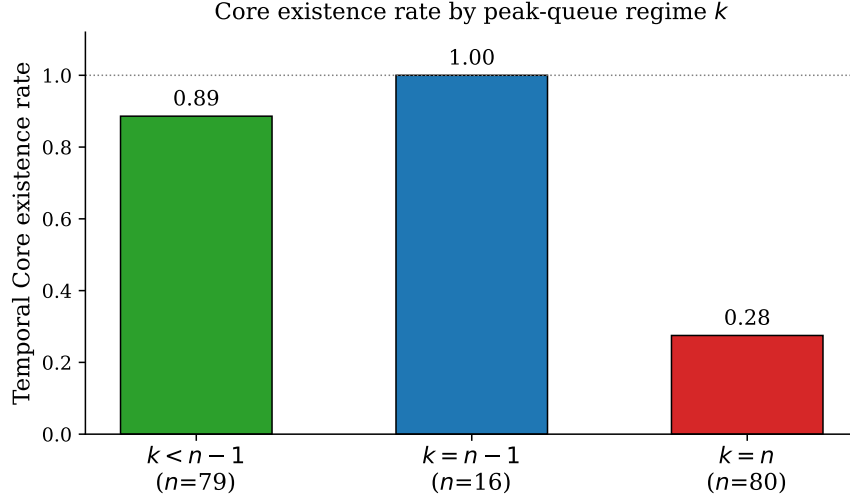


Figure 2: Temporal Core existence rate stratified by peak-queue regime k . The complement-coalition mechanisms (Theorem 9, Proposition 10, Proposition 13) all require $k \geq n-1$. Empirically, 58 of the 67 empty Cores concentrate at $k = n$ via single / balanced / balanced-near-complement / near-complement-residual mechanisms; the remaining 9 occur at $k < n-1$ via the intermediate-coalition mechanism of Observation 15, all in patterns B_{medium} or E.

nonempty in all 25 instances for each pattern. In pattern E the same obstruction blocks complement mechanisms, yet 2 of the 25 instances still exhibit empty Cores via the intermediate-coalition mechanism of Observation 15—see §6.3. The operational content of Corollary 18 is therefore precise: a shallow queue rules out complement and balanced-complement mechanisms, but does not by itself prevent intermediate-coalition violations at high $\bar{r}$.

Load parameter ρ alone does not determine Core stability. Patterns B_{medium} and C share the same load parameter $\rho = 2$ yet exhibit qualitatively different behavior. B_{medium} (uniform sequential arrivals) is complement-mechanism-free in most instances (Theorem 9 is applicable in only 13/25 and fires in 3/25), yet 7 further instances are empty through the intermediate mechanism, giving an overall Core rate of 0.56. C (two angularly interleaved batches) places almost every instance (24/25) into the Theorem 9-applicable regime, where single-complement violations fire in 10/25, producing a Core rate of 0.28. The structural difference in the arrival geometry alone accounts for this gap. This confirms that ρ , while necessary for placing a pattern on the saturation spectrum, is not sufficient to characterize Core stability: the spatial-temporal structure of the arrival process and the dispatch-policy dynamics both matter independently.

6.5 Sharpness of r^{**} versus r^*

Empirically, on the NN grid $\bar{r}^{**} = 1.223$ and $\bar{r}^{***} = 1.096$ versus $\bar{r}^* = 4.351$; the sharpness factors are $r^*/r^{**} \approx 3.56$ and $r^*/r^{***} \approx 3.97$. Because the realized ratio $\bar{r} = 1.312$ is close to r^{**} and r^{***} but far below r^* , Theorem 9 and Proposition 10 are operationally informative whereas

Table 6: Pattern-level aggregate statistics (NN policy, $5 \times 5 = 25$ instances per pattern; overall row pools all 175 instances). $\bar{r}^*$, being a function of coordinates only ($r^* = \sum_i c(\{i\})/c^*(N)$), is identical across all patterns at 4.351 and reported as an overall value rather than per-pattern. Em-dashes in $\bar{r}^{**}$ (resp. $\bar{r}^{***}$) indicate patterns where Theorem 9 (resp. Proposition 10) is never applicable. Bracketed quantities in the Core-rate column are Wilson 95% confidence intervals; per-pattern intervals have full width ≈ 30 –35 percentage points (half-width ± 15 –18 pp) reflecting the $n = 25$ replicate count, and the overall interval narrows to ≈ 14 pp full width (± 7 pp).

Pattern	ρ	Core rate [95% CI]	$\bar{r}$ (sd)	$\bar{r}^{**}$ (sd)	$\bar{r}^{***}$	$ \mathcal{F} /(2^n - 1)$	Thm 9 app/fires	Prop 10 app/fires
A	∞	0.36 [0.20, 0.55]	1.125 (0.102)	1.181 (0.059)	1.092	1.000	25 / 9	25 / 15
B _{heavy}	4.0	0.20 [0.09, 0.39]	1.233 (0.127)	1.187 (0.061)	1.094	0.985	25 / 15	24 / 20
B _{medium}	2.0	0.56 [0.37, 0.73]	1.273 (0.135)	1.305 (0.148)	1.150	0.590	13 / 3	7 / 4
B _{light}	0.5	1.00 [0.87, 1.00]	1.690 (0.210)	—	—	0.077	0 / 0	0 / 0
C	2.0	0.28 [0.14, 0.48]	1.197 (0.149)	1.181 (0.059)	1.087	0.985	25 / 10	24 / 18
D	1.0	1.00 [0.87, 1.00]	1.332 (0.121)	1.398 (0.217)	—	0.196	3 / 0	0 / 0
E	1.0	0.92 [0.75, 0.98]	1.337 (0.169)	1.505 (0.139)	—	0.273	5 / 0	0 / 0
Overall	—	0.62 [0.54, 0.69]	1.312 (0.224)	1.223 (0.121)	1.096	0.587	96 / 37	80 / 57

Remark 5 almost never fires. The classical bound grows with n because $r^* = \sum_i c(\{i\})/c^*(N)$ has a numerator scaling linearly in n while $c^*(N)$ scales like $\sqrt{n}$ on uniform planar instances; r^{**} and r^{***} , in contrast, are pinned near 1 because complement coalitions $N \setminus \{i\}$ nearly partition the grand tour. This explains why the gap between the two widens with n and quantifies the practical advance of the temporal analysis. Supplementary §S10 (Supplementary Figure S2) plots the three threshold distributions side by side.

6.6 Robustness to dispatch policy

The three sufficient conditions of §5 are policy-independent in their threshold values—which depend only on TSP costs and the arrival sequence, not on service times—and applicability counts differ by at most one across nearest-neighbor (NN), cheapest-insertion (CI), and batch re-optimization (BR) dispatch over $175 \times 3 = 525$ policy-instance pairs; all three thresholds preserve the no-false-positives guarantee, while the intermediate-coalition mechanism of Observation 15—observed only under NN—disappears under both CI and BR. Detailed per-policy results, including the dispatch-policy contingency table and the empty-mechanism breakdown, appear in Supplementary §S5.

6.7 Scale-up to larger n

Our main study at $n \leq 15$ is capped by the $O(n^2 2^n)$ cost of Held–Karp. To test whether the theoretical and empirical findings of the preceding subsections persist at larger scales, we extend the study to $n \in \{20, 30, 50\}$ for three representative patterns—A ($\rho = \infty$, simultaneous), B_{medium} ($\rho = 2$, sequential), and C ($\rho = 2$, clustered)—spanning the Core-fragile, Core-safe, and intermediate regimes. Each cell uses 5 seeds, giving 45 scale-up instances with NN dispatch.

Solver and calibration follow the methodology described in Supplementary §S3.

Four findings emerge.

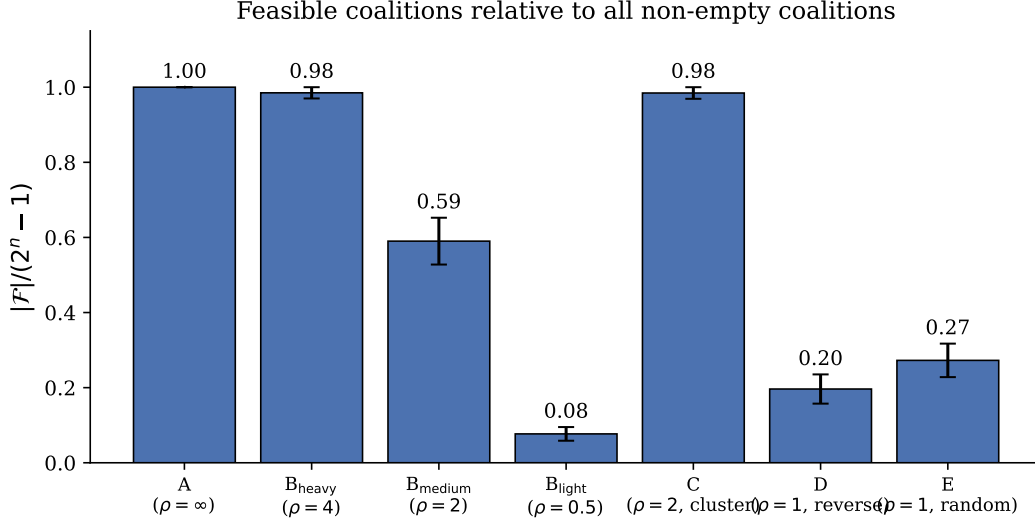


Figure 3: Relative size of the feasible coalition family, $|F|/(2^n - 1)$.

Finding 1: $\bar{r}^{}$ decays with n , confirming Theorem 16.** For pattern A, $\bar{r}^{**}$ drops from 1.181 at $n = 15$ (Table 6) to 1.119, 1.096, 1.068 at $n = 20, 30, 50$, respectively. Pattern C exhibits the same decay (B_{medium} has no feasible complement at any scale, so $\bar{r}^{**}$ is undefined). The observed rate of decrease is consistent with the $O(n^{-1/2})$ bound of Theorem 16 (and within the $O(n^{-1})$ improvement of Remark 17 for the simultaneous-arrival case A). A log-log regression of $\bar{r}^{**} - 1$ on n for Pattern A yields slope -0.54 ($R^2 = 0.99$, eight per- n mean points $n \in \{5, 7, 10, 12, 15, 20, 30, 50\}$) and -0.62 ($R^2 = 1.00$, four per- n mean points on the scale-up subset $n \in \{15, 20, 30, 50\}$), close to Theorem 16’s $-1/2$ and partway toward Remark 17’s -1 .

Finding 2: Pattern C activates the complement-mechanism as n grows. Pooled over $n \in \{5, 7, 10, 12, 15\}$, pattern C has $\bar{r} = 1.197$ marginally above $\bar{r}^{**} = 1.181$ (Table 6), with a 28% Core-existence rate overall; at $n = 15$ under NN all 5 instances are already empty (mean $r = 1.238$ vs. mean $r^{**} = 1.145$). By $n = 20$ the gap widens substantially ($\bar{r} = 1.335$ vs. $\bar{r}^{**} = 1.119$), and all 5 test instances are empty; the same holds at $n = 30, 50$. Two concurrent effects are at work: $\bar{r}$ increases (the NN competitive ratio degrades with more customers and the clustered zig-zag accumulates realized waiting), while $\bar{r}^{**}$ decreases by Theorem 16. Clustered patterns thus shift from marginal to strong complement-mechanism activation as n grows.

Finding 3: Pattern B_{medium} blocks the complement mechanism structurally at every scale; intermediate-coalition violations are not detected in the sampled LP but cannot be ruled out. We split this finding into an analytically sharp structural claim (a) and a one-sided empirical observation (b) whose detection power decreases with n .

(a) *Structural claim (analytic, strong).* Across $n \in \{20, 30, 50\}$, B_{medium}’s sparse arrivals keep $\bar{k}$ strictly below $n - 1$ ($\bar{k} = 15.4, 23.2, 37.6$), so every complement $N \setminus \{i\}$ is infeasible and all three

Table 7: Scale-up aggregate statistics (5 seeds each). Em-dashes in $\bar{r}^{**}$ indicate no feasible complement coalition (Theorem 9 vacuous). For pattern A with $n \geq 30$ the Core LP is intractable to enumerate, since $|\mathcal{F}| = 2^n - 1$; emptiness there is certified by Theorem 9 (conditional on LKH-computed $c^*(S)$ values for $n > 20$; minimum $r - r^{**}$ margin 0.0159 in absolute ratio units; this is several orders of magnitude larger than the relative LKH-vs-Held-Karp calibration error of 0.000% measured at $n = 15$, so the analytic certification is not threatened by heuristic-solver inexactness) for the 4 of 5 seeds at $n = 30$ and 5 of 5 at $n = 50$ where $r > r^{**}$, and by a restricted Core LP (Supplementary Materials §S3) for the one seed at $n = 30$ where $r \leq r^{**}$. B_{medium} rows report a one-sided sampled-LP observation only, not a proof of full Core nonemptiness (see footnote § and Finding 3(b)).

Pattern	n	Core rate (full/analytic)	No-viol. rate (sampled LP) [§]	$\bar{r}$	$\bar{r}^{**}$	$\bar{k}$
A	20	0 / 5 [†]	—	1.167	1.119	20.0
A	30	0 / 5 ^{†‡}	—	1.187	1.096	30.0
A	50	0 / 5 [†]	—	1.211	1.068	50.0
B_{medium}	20	—	5 / 5 [§]	1.330	—	15.4
B_{medium}	30	—	5 / 5 [§]	1.309	—	23.2
B_{medium}	50	—	5 / 5 [§]	1.387	—	37.6
C	20	0 / 5	—	1.335	1.119	20.0
C	30	0 / 5	—	1.282	1.096	30.0
C	50	0 / 5	—	1.282	1.068	50.0

[†] Core LP not enumerated at this scale; Theorem 9 fires in 4/5 seeds at $n = 30$ and 5/5 at $n = 50$; the one non-firing seed at $n = 30$ (seed 123) is certified by a restricted Core LP (Supplementary Materials §S3). [‡] At $n = 20$, Theorem 9 is vacuous ($r \leq r^{**}$) at seeds 99 and 123 and fires at seeds 7, 42, 256; at $n = 30$, it is vacuous at seed 123 only. Core emptiness for the vacuous seeds is certified by the near-complement mechanism (Remark 12, Section 6.3). [§] No intermediate-coalition violation detected in the sampled restricted LP of 10^4 coalitions per instance; this does *not* certify full Core nonemptiness (per-draw detection fraction $\approx 10^{-2}, 10^{-5}, 10^{-11}$ at $n = 20, 30, 50$ respectively). The structural statement (complement mechanisms blocked by Corollary 18) is analytically sharp; the no-violation observation is one-sided. At $n > 15$ the Core-existence judgment is made via the restricted LP of Supplementary Materials §S3 rather than full $2^n - 1$ enumeration.

complement-based mechanisms (Theorem 9, Proposition 10, and Proposition 13) are structurally blocked by Corollary 18, regardless of the realized r .

(b) *Empirical observation (one-sided, weak; see detection-limit warning below).* Across the 15 instances at $n \in \{20, 30, 50\}$, $\bar{r} \in [1.31, 1.39]$; no intermediate-coalition violation is observed in our sampled restricted-LP of $K = 10^4$ coalitions per instance. This is a one-sided certificate only: the sampled LP cannot conclude full Core nonemptiness, and the sampling fraction $K/(2^n - 1)$ collapses as n grows ($\approx 10^{-2}, 10^{-5}, 10^{-11}$ at $n = 20, 30, 50$). Absence of observed violation at $n = 50$ is therefore not evidence of rarity.

Detection-limit warning ($n = 50$ specifically). At $n = 50$ the per-draw detection probability for any single binding intermediate-size coalition is $\sim 10^{-11}$, so the absence of violations in our $K = 10^4$ sampled LP carries essentially no evidential weight: Finding 3(b) at $n = 50$ should be read as a placeholder for “intractable to test.” The caveat is weaker at $n = 30$ ($\sim 10^{-5}$ per draw, joint detection still under 10%) and operationally negligible at $n = 20$. Only the structural claim

3(a) carries asymptotic strength.

Finding 4: Near-complement mechanism at seed 123 across scales. Seed 123 under Pattern A exhibits a two-tier certification pattern consistent with Theorem 16’s $O(n^{-1/2})$ tightening of $\bar{r}^{**}$:

- At $n = 20$ and $n = 30$, Theorem 9 is vacuous ($r = 1.077 < r^{**} = 1.112$ and $r = 1.038 < r^{**} = 1.088$ respectively). A restricted Core LP over 10,000 sampled coalitions certifies Core emptiness with binding constraints concentrated at sizes $n - 1$ and $n - 2$ simultaneously, consistent with the near-complement mechanism of Remark 12. These three cases constitute the Supplementary Materials §S3 case study.
- At $n = 50$, Theorem 9 directly fires ($r = 1.306 > r^{**} = 1.081$), analytically certifying Core emptiness without requiring restricted-LP evidence. The threshold r^{**} at $n = 50$ is small enough (by Theorem 16’s decay) that the realized competitive ratio clears it, so the analytic certificate takes over and the near-complement auxiliary no longer needs to be invoked.

By contrast, a clean Theorem-9-fires instance at smaller scale (Pattern A, $n = 30$, seed 42) binds only at size $n - 1$ (Section 6.3). This transition from restricted-LP certification at the smaller seed-123 scales ($n = 20, 30$) to analytical certification at $n = 50$ is the natural consequence of $r^{**} \rightarrow 1$; seed 123 is not itself anomalous in the asymptotic limit, only uncovered by Theorem 9 at finite n below the transition.

Intractability of pattern A at $n \geq 30$. For pattern A, $|\mathcal{F}| = 2^n - 1$, so the Core LP has $\sim 10^9$ and $\sim 10^{15}$ constraints at $n = 30$ and $n = 50$ respectively, well beyond enumeration. Theorem 9 nevertheless certifies Core emptiness *a priori* in 12 of the 15 seed–size combinations: every complement coalition is feasible, so r^{**} is well-defined, and numerically $r > r^{**}$ holds in 3/5 seeds at $n = 20$, 4/5 at $n = 30$, and 5/5 at $n = 50$ (Table 7). The three exceptions—seeds 99 and 123 at $n = 20$, and seed 123 at $n = 30$ —have $r \leq r^{**}$; for these the restricted Core LP of Supplementary Materials §S3 certifies emptiness (Finding 4 above). The intractability of pattern A therefore itself *demonstrates* the $k < n - 1$ versus $k = n$ divide identified in Corollary 18: the regime where the Core LP becomes infeasible to enumerate is precisely the regime where Theorem 9—supplemented by the near-complement certificate at the few exceptional seeds—can certify emptiness without enumeration.

Summary. The qualitative findings persist to $n = 50$: the Core-empty tail at $n = 15$ (pattern C) becomes near-certain by $n = 20$ via r^{**} tightening (Theorem 16), while Corollary 18’s structural safeguard only strengthens ($\bar{k} < n - 1$ analytically blocks complement mechanisms at every scale-up cell). The empirical companion for B_{medium} —no intermediate violation observed in our sampled LP—is one-sided and detection-power limited (Finding 3(b)); it does not bound the probability of empty Core. The reading is: one strong analytic conclusion (structural blocking) plus one weak

empirical one (no sampled violation), consistent with the main-grid observation that intermediate empties vanish under CI/BR (§6.6).

7 Conclusion

We introduced the Online Traveling Salesman Game and defined the Temporal Nucleolus as the lexicographic excess minimizer over the restricted family $\mathcal{F}$ of coalitions realizable under the online arrival process. The classical (static) TSG is recovered at the boundary case of simultaneous arrivals; the Temporal Core is empty whenever the competitive ratio exceeds any of three complement-based thresholds (Theorem 9, Proposition 10, Proposition 13) or the unifying partition-pair certificate of Proposition 14; a bounded waiting queue ($k < n - 1$) structurally blocks the three complement-based mechanisms (Corollary 18), but not the partition-pair mechanism via non-singleton splits. On the 175-instance NN grid, the four analytic thresholds jointly certify 66 of 67 observed empty Cores with no false positives—1 near-complement residual completes the five-mechanism decomposition. The thresholds are sharper than the classical bound by $3.56\times$ ($\bar{r}^{**} = 1.223$) and $3.97\times$ ($\bar{r}^{***} = 1.096$), making them the first operationally informative thresholds for cost-sharing stability in online routing.

Platform-design implication. From a platform-design perspective, our results offer suggestive evidence that enforcing short service-time promises—equivalently, keeping the peak queue depth below $n - 1$ —structurally rules out all three complement-coalition mechanisms for emptying the Core. In the shallow-queue regime, the partition-pair mechanism of Proposition 14 remains logically possible (Observation 15), and is empirically observed only under nearest-neighbor dispatch: cheapest-insertion and batch re-optimization produce zero such cases in our grid, suggesting that even moderately informed dispatch policies mitigate this mechanism (Section 6.6).

Operational trade-offs. The platform-design recommendation that bounding $k < n - 1$ structurally blocks the complement-coalition mechanisms (Corollary 18) is not without operational cost. Achieving $k < n - 1$ in practice requires either tighter service-time commitments (raising delivery-speed standards), increased fleet capacity (more vehicles serving simultaneously), or more frequent dispatch (reducing idle waiting at the depot); each imposes resource costs that the cooperative-game analysis here does not quantify. Moreover, k is itself an endogenous outcome of the arrival rate and service rate distributions, not a free control variable: a platform with high request volume may face $k = n$ regimes even with aggressive dispatching. Whether the Core-stability gain from bounding k is worth the operational cost is a platform-design judgment that depends on cost-of-defection considerations not modeled here. Combining the cooperative game with explicit fleet-capacity costs along the lines of Özener & Ergun (2008) would clarify this trade-off.

Limitations and future work. Five concerns merit explicit comment.

- (i) *Dispatch policy.* Theorem 9, Supplementary Proposition 3 (§S7), and Corollary 18 use only the realized $(r, \mathcal{F}, k)$, hence are policy-independent; Section 6.6 verifies this empirically across NN/CI/BR. Learning-based or forecasting-aware dispatch remains open.
- (ii) *Instance scale.* Section 6.7 extends to $n \in \{20, 30, 50\}$ with LKH for $c^*(S)$ at $n > 20$ (calibration error 0.000% vs. Held–Karp at $n = 15$, well below the minimum $r - r^{**}$ margin of 0.0159). For B_{medium} the complement mechanisms are structurally blocked at every scale (Corollary 18); the intermediate-coalition mechanism is not flagged by our $K = 10^4$ sampled LP, but the sampling fraction $K/(2^n - 1)$ collapses with n , so the B_{medium} nonemptiness claim is structural plus one-sided empirical, not a proof. Extending to $n \geq 100$, non-uniform geometries, heavy-tailed queueing, and full- $\mathcal{F}$ scale-up certification is open.
- (iii) *Near-complement residual.* Proposition 13 closes 9 of 10 main-grid cases; the residual ($n = 15$, A, seed 42) misses by $\approx 2 \times 10^{-3}$. Whether enlarging the balancing family to $B_{n-1} \cup B_{n-2} \cup B_{n-3}$ or relaxing the efficiency equality to a covering inequality closes this gap is open (Supplementary §S4 records the diagnostic).
- (iv) *Intermediate-coalition mechanism.* Proposition 14 certifies the 9 cases of Observation 15 via dual support extraction (Supplementary §S9). What remains open is a policy-aware predictive condition derived from queueing primitives (queue-depth distribution, service-time variability) rather than after-the-fact LP duals. The mechanism is also policy-sensitive: observed only under NN, absent under CI/BR.
- (v) *Policy-dependence of $\mathcal{F}$.* Threshold values $(r^{**}, r^{***}, r_{\min}^{(\diamond)}, r_{\min}^{(P)})$ are policy-independent; the family $\mathcal{F}$ itself depends on realized service times and hence on the dispatch policy. The policy-induced framing adopted here is operationally realistic; a counterfactual framing in which $\mathcal{F}$ depends only on arrival times is more robust to policy changes. Systematic comparison is left for future work.

Scope demarcation and Online VRG. The single-vehicle scope is a deliberate methodological choice: it isolates the temporal coalition-feasibility structure of $\mathcal{F}$ from fleet-assignment confounds, and the three complement-based sufficient conditions rely on $c(S)$ being a single TSP tour cost for which the insertion identity $c(\{i\}) + c(N \setminus \{i\}) \geq c^*(N)$ holds. An *Online Vehicle Routing Game* combining $\mathcal{F}$ with multi-vehicle capacity (Göthe-Lundgren et al., 1996; Engevall et al., 2004) requires redefining $c(S)$ jointly with capacity and a fleet-assignment rule, under which that insertion identity can fail; this opens a separate agenda (capacitated coalition costs, fleet-level balancing, A3-style rolling-horizon consistency (Lehrer & Scarsini, 2013; Kimms & Kozeletskyi, 2016b)) that a companion manuscript will develop.

Data and code availability

All code, data, and reproducibility scripts are publicly available at <https://github.com/sage-Tae/online-tsg-2026>.

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